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PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL

Abstract

A packing assembly includes a packing sleeve having a first sleeve portion and a second sleeve portion, wherein a first sleeve portion outer diameter is larger than a second sleeve portion outer diameter and a first sleeve portion inner diameter is larger than a second sleeve portion inner diameter. The packing assembly also includes a groove formed in the second sleeve portion outer diameter. The packing assembly further includes a shoulder formed at a transition between the first sleeve portion and the second sleeve portion. The packing assembly includes a shelf formed at the transition between the first sleeve portion and the second sleeve portion. The packing assembly also includes a plurality of packing components arranged within the first sleeve portion inner diameter, at least a portion of the plurality of packing components engaging the shelf. The packing assembly further includes a seal positioned within the groove.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 17/890,818, filed Aug. 18, 2022, titled “PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL,” which is a divisional of U.S. patent application Ser. No. 17/308,163, filed May 5, 2021, titled “PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL,” now U.S. Pat. No. 11,421,679 issued Aug. 23, 2022, which is a continuation-in-part of U.S. patent application Ser. No. 16/916,657, filed Jun. 30, 2020, titled “SLEEVED PACKING BORE SEALING SYSTEM,” the full disclosures of which are hereby incorporated by reference in their entirety for all purposes.

TECHNICAL FIELD

[0002] Embodiments of the subject matter disclosed herein generally relate to pump systems, and in particular to sealing systems.

BACKGROUND

[0003] Pumping systems may be used in a variety of applications, such as industrial applications where pumping systems are used to elevate a working fluid pressure. One such application is hydraulic fracturing systems, where high pressure pumps are used to increase a fluid pressure of a working fluid (e.g., fracturing fluid, slurry, etc.) for injection into an underground formation. The working fluid may include particulates, which are injected into fissures of the formation. When the fluid is removed from the formation, the particulates remain and “prop” open the fissures, facilitating flow of oil and gas. In many applications, reciprocating pumps are used where a fluid is introduced into a fluid end inlet passage and out through an outlet passage. A plunger reciprocates within a bore to add energy to the fluid. Due to the particulates and corrosive nature of the working fluid, various sealing surfaces become eroded or otherwise damaged.

SUMMARY

[0004] Applicants recognized the problems noted above herein and conceived and developed embodiments of systems and methods, according to the present disclosure, for sealing systems.

[0005] In accordance with one or more embodiments, a fluid end includes a first bore extending through a body toward a pressure chamber. The fluid end also includes a second bore extending through the body and intersecting the pressure chamber. The fluid end further includes a packing assembly positioned within the first bore. The packing assembly includes a seal positioned against an inner face of the first bore. The packing assembly also includes a sleeve positioned against the seal, both the sleeve and the seal arranged at a radially outward location of the first bore, the sleeve having radially outwardly facing coupling components at an end opposite the seal. The packing assembly further includes a ring positioned radially inward of the seal. The packing assembly

includes a plurality of packing components positioned on the ring.

[0006] In accordance with another embodiment, a packing assembly includes a packing sleeve having a first end and a second end, the first end having a coupling component and configured to extend, at least partially, axially out of bore of a fluid end block. The packing assembly also includes a seal adapted to sit on a shelf formed within the bore, the seal positioned for engagement by the second end of the packing sleeve. The packing assembly further includes a ring configured to sit on the shelf, radially inward of the seal, at least a portion of the ring axially overlapping the second end along the bore. The packing assembly includes a plurality of packing components configured to sit on a ring shelf, the plurality of packing components to be positioned radially inward of the sleeve and the seal.

[0007] In accordance with another embodiment, a method includes threadingly engaging a first sleeve arranged within a bore of a pump assembly via a tool. The method also includes coupling the tool to the pump assembly. The method further includes axially removing the first sleeve from the bore, responsive to at least partial first operation of the pump assembly. The method also includes threadingly engaging a second sleeve to the tool. The method includes axially installing the second sleeve into the bore, responsive to at least partial second operation of the pump assembly.

[0008] In accordance with another embodiment, a fluid end includes a first bore extending through a body toward a pressure chamber. The fluid end also includes a second bore extending through the body and intersecting the pressure chamber. The fluid end further includes a packing assembly positioned within the first bore, the packing assembly having a sleeve that compresses a seal against at least a portion of the first bore to block fluid flow from the first bore, the sleeve extending along a sleeve length, wherein at least a first portion of the sleeve length is positioned within the first bore and at least a second portion of the sleeve length extends axially beyond the first bore, the second portion having a coupling component, and packing components positioned radially inward at least a portion of the sleeve. The fluid end also includes a cap coupled to the body, the cap overlapping the first portion of the sleeve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The present technology will be better understood on reading the following detailed description of non-limiting embodiments thereof, and on examining the accompanying drawings, in which:

[0010] FIG. 1 is a schematic cross-sectional view of an embodiment of a pump assembly, in accordance with embodiments of the present disclosure;

[0011] FIG. 2A is a schematic cross-sectional view of an embodiment of a pump assembly including a packing assembly, in accordance with embodiments of the present disclosure;

[0012] FIG. 2B is a detailed cross-sectional view of an embodiment of a pump assembly including a packing assembly, in accordance with embodiments of the present disclosure;

[0013] FIG. 3A is a schematic cross-sectional view of an embodiment of a packing assembly, in accordance with embodiments of the present disclosure;

[0014] FIG. 3B is a detailed cross-sectional view of an embodiment of a packing assembly, in accordance with embodiments of the present disclosure;

[0015] FIG. 3C is a detailed cross-sectional view of an embodiment of a packing assembly, in accordance with embodiments of the present disclosure;

[0016] FIG. 4 is an exploded cross-sectional view of an embodiment of a pump assembly, in accordance with embodiments of the present disclosure;

[0017] FIG. 5 is a cross-sectional view of an embodiment of a sleeve, in accordance with

embodiments of the present disclosure;

[0018] FIGS. 6A-6F are cross-sectional views of an embodiment of a sequence for removal of a sleeve, in accordance with embodiments of the present disclosure;

[0019] FIGS. 7A-7C are isometric views of an embodiment of a sequence for removal of a sleeve, in accordance with embodiments of the present disclosure;

[0020] FIG. 8 is a schematic cross-sectional view of an embodiment of a pump assembly including a packing assembly, in accordance with embodiments of the present disclosure;

[0021] FIG. 9A is a detailed cross-sectional view of an embodiment of a pump assembly including a packing assembly, in accordance with embodiments of the present disclosure;

[0022] FIG. 9B is a detailed cross-sectional view of an embodiment of a pump assembly including a packing assembly, in accordance with embodiments of the present disclosure;

[0023] FIG. 10 is an exploded view of an embodiment of a pump assembly, in accordance with embodiments of the present disclosure;

[0024] FIG. 11 is a cross-sectional view of an embodiment of a sleeve coupled to a tool, in accordance with embodiments of the present disclosure; and

[0025] FIG. 12 is a flow chart of an embodiment of a method for sleeve installation and removal, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0026] The foregoing aspects, features, and advantages of the present disclosure will be further appreciated when considered with reference to the following description of embodiments and accompanying drawings. In describing the embodiments of the disclosure illustrated in the appended drawings, specific terminology will be used for the sake of clarity. However, the disclosure is not intended to be limited to the specific terms used, and it is to be understood that each specific term includes equivalents that operate in a similar manner to accomplish a similar purpose.

[0027] When introducing elements of various embodiments of the present disclosure, the articles “a”, “an”, “the”, and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including”, and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Any examples of operating parameters and/or environmental conditions are not exclusive of other parameters/conditions of the disclosed embodiments. Additionally, it should be understood that references to “one embodiment”, “an embodiment”, “certain embodiments”, or “other embodiments” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, reference to terms such as “above”, “below”, “upper”, “lower”, “side”, “front”, “back”, or other terms regarding orientation or direction are made with reference to the illustrated embodiments and are not intended to be limiting or exclude other orientations or directions.

[0028] Embodiments of the present disclosure are directed to one or more sealing systems for pumps, which may include reciprocating pumps. In at least one embodiment, one or more sealing systems may further be incorporated in various systems for pressure containment, and it should be appreciated that descriptions with respect to pumping systems or pumps is by way of example only and not intended to limit the scope of the present disclosure. Embodiments may be directed toward a sleeve (e.g., sleeve assembly, packing sleeve, insert, etc.) that includes external threads positioned for engagement by an installation/retrieval tool. In at least one embodiment, external threads are not engaged with other components of the pumping system during normal pumping operations. Accordingly, systems and methods may be directed toward a sleeve that interacts with an installation tool to facilitate rapid installation and removal of one or more components of a sealing system.

[0029] In at least one embodiment, systems and methods may be directed toward an installation tool that threadingly engages one or more sleeves, which may include mating threads as noted

above. The installation tool may further be coupled to one or more pump pony rods via a clamp or other reasonable connection. In at least one embodiment, the installation tool may be used for rapid removal and installation of one or more sealing systems. By way of example only, a sleeve may be removed by threading or otherwise coupling a tool to the sleeve and engaging the tool to the pump, as noted above. Driving rotation of the pump crankshaft may facilitate removal of the sleeve. Thereafter, the sleeve may be threaded and a replacement may be coupled to the tool. Subsequent rotation of the crankshaft in an opposite direction may then be used to install the replacement sleeve.

[0030] In at least one embodiment, the sleeve may, at least in part secure a portion of packing or other sealing material within the a pump bore. For example, the seal may be positioned axially next to the packing and installation may compress the sleeve against the packing. In various embodiments, the sleeve may include one or more shelves to receive at least a portion of the packing material. In at least one embodiment, the sleeve may include an extension or arm to compress or otherwise engage packing material. In at least one embodiment, the sleeve may include combinations of elements to enable installation in a variety of different configurations. By way of example only, the sleeve may include a shelf, but the shelf may not be used in configurations where the sleeve is used to axially compress the packing. Moreover, the shelf may be used as the extension and not to receive the packing material.

[0031] FIG. 1 is a schematic cross-sectional view of an embodiment of a pump assembly **100**, which may also be referred to as a reciprocating pump assembly and/or a reciprocating pump. The pump assembly **100** may be utilized during hydraulic fracturing operations, among other operations, where a working fluid (e.g., fracturing fluid, slurry, etc.) is introduced into the pump and energy is added to the working fluid to increase a pressure of the working fluid. Fracturing fluid, by way of example only, may include corrosives and also particulates, such as sand or ceramics, which are utilized during fracturing operations. These corrosives and particulates cause erosion within the pump assembly **100**, which may undesirably affect fracturing operations and lead to down times to replace various components. Additionally, the fracturing fluids may include corrosive acids and the like, which may wear down components of the pump assembly **100**.

[0032] It should be appreciated that various components of the pump assembly **100** have been removed for clarity with the following discussion. For example, a power end has been removed in favor of focusing on the illustrated fluid end **102** of the pump assembly **100**. The power end may include a crankshaft that is driven by an engine or motor to facilitate operations. The fluid end **102** includes a fluid end block **104** that may house one or more components discussed herein. A plunger rod **106** is driven (e.g., via the crankshaft) to reciprocate within the fluid end block **104** along a plunger axis **108**. The plunger rod **106** is positioned within a bore **110** extending through at least a portion of the fluid end block **104**. The illustrated bore **110** is arranged along the plunger axis **108** (e.g., first axis) and intersects a pressure chamber **112**, which is arranged along a pressure chamber axis **114** (e.g., second axis), which is positioned substantially perpendicular to the plunger axis **108**. It should be appreciated that the pump assembly **100** may include multiple plunger rod and pressure chamber arrangements, which may be referred to as a plunger throw. For example, the pump assembly **100** may be a triplex pump, quadplex pump, quintuplex pump, and the like.

[0033] The illustrated fluid end block **104** includes an inlet passage **116** and an outlet chamber **118**, which are generally coaxial and arranged along the pressure chamber axis **114**. In other words, the inlet passage **116** and the outlet chamber **118** are axially aligned with respect to one another and/or the pressure chamber **112**. In various embodiments, fluid enters the pressure chamber **112** via the inlet passage **116**, for example on an up stroke of the plunger rod **106**, and is driven out of the pressure chamber **112** an outlet passage **120**, for example on a down stroke of the plunger **106**.

[0034] Respective valve assemblies **122**, **124** are arranged within the inlet passage **116** and the outlet chamber **118**. These valve assemblies **122**, **124** are spring loaded in the illustrated embodiment, but it should be appreciated that such an arrangement is for illustrative purposes only.

In operation, a differential pressure may drive movement of the valve assemblies. For example, as the plunger rod **106** is on the upstroke, pressure at the inlet passage **116** may overcome the spring force of the valve assembly **122**, thereby driving fluid into the pressure chamber **112**. However, on the down stroke, the valve assembly **122** may be driven to a closed position, while the spring force of the valve assembly **124** is overcome, thereby enabling the fluid to exit via the outlet passage **120**.

[0035] In one or more embodiments, a sealing element (not pictured) is arranged about the plunger rod **106** to block high pressure fluid from flow outside of the block **104**. In at least one embodiment, one or more shoulders or groove may be formed within the block **104** to receive at least a portion of the sealing elements or components associated with the sealing elements. By way of example, one or more sealing elements may be positioned against a shoulder or groove within a bore and then compressed and/or axially maintained at a location by a sleeve that is secured to the block **104**, such as by fasteners, threads, or the like. In at least one embodiment, installation and removal of the sleeve may be a challenging or time consuming process, and accordingly, systems and methods are directed toward sleeves and/or installation/removal tools to facilitate installation and removal of sealing elements.

[0036] FIGS. 2A and 2B are cross-sectional views of the pump assembly **100** where the plunger rod **106** is illustrated within the bore **110**. It should be appreciated that the configuration of various components of FIGS. 2A and 2B are by way of example only and are not intended to limit the scope of the present disclosure. As will be described herein, an illustrated sleeve may include a shelf in at least one embodiment, but the shelf may be omitted in other embodiments, as noted herein and further described below. The illustrated bore **110** includes a variable diameter including a stepped portion, which will be described herein, and receives a packing assembly **200**, which includes a packing sleeve **202** (e.g., sleeve). The sleeve **202** forms a barrier between a wall **204** of the bore **110** and the plunger rod **106**. As will be described below, during operation the sleeve **202** may be subjected to harsh operating conditions and erosion. It may be desirable to form the sleeve **202** as a sacrificial component for replacement over a lifetime of the pump assembly **100**. For example, the sleeve **202** may be formed from a different material than the block **104** and/or from a material with altered/enhanced corrosion or erosion resistant properties.

[0037] The illustrated packing assembly **200** is positioned within the bore **110** that includes a first bore section **206**, a second bore section **208**, and a third bore section **210**. It should be appreciated that the three sections **206**, **208**, **210** are shown for illustrative purposes and that in other embodiments more or fewer sections may be included. The illustrated bore **110** has a variable diameter across an axial length **212** extending from a face **214** toward the pressure chamber **112**. In this configuration, a bore diameter decreases from the face **214** to the chamber **112**. For example, the first bore section **206** has a first bore diameter **216**, the second bore section **208** has a second bore diameter **218**, and the third bore section **210** has a third bore diameter **220**. The first bore diameter **216** is larger than both the second bore diameter **218** and the third bore diameter **220**, while the second bore diameter **218** is also larger than the third bore diameter **220**. The variable diameter bore may enable a shelf or shoulder to receive at least a portion of the sleeve **202**, which may facilitate locating the sleeve **202** in a desired position and/or provide an indication to an operator for successful installation, among other benefits.

[0038] The sleeve **202** is a substantially cylindrical component having a body **222** with a bore **224** extending through the body **222**. The bore **224** is a variable diameter bore **224** and the body **222** includes a first sleeve portion **226** and a second sleeve portion **228**, each having a respective diameter **230**, **232** (e.g., inner diameter). As noted above, and will be described herein, other embodiments may include a constant or substantially constant diameter bore **224** with both the first sleeve portion **226** and the second sleeve portion **228** having a common diameter **230**, or in some embodiments, a common diameter **232**. In this embodiment, the first sleeve portion diameter **230** is larger than the second sleeve portion diameter **232**. As illustrated, the change in diameters forms a

shelf **234** for receiving one or more packing components **236**, such as seals, packing brass, and the like. In at least one embodiment, the shelf **234** is omitted, for example where the diameter is constant. With the inclusion of the packing components **236**, a working diameter of the first sleeve portion is reduced, such that the packing component diameter **238** may be substantially equal to the second sleeve portion diameter **232**. In operation, the plunger **106** reciprocates through the sleeve **202** such that the packing components **236** may contact the plunger **106**.

[0039] As mentioned, in various embodiments the sleeve **202** is arranged within the bore **110** to provide a sacrificial component between the block **104** and the plunger **106**. To prevent leaks and/or significantly reduce a likelihood of leaks, there may be seals and the like arranged between the sleeve **202** and the bore **110**. In this embodiment, the sleeve **202** includes a groove **240** that receives seal **242** in the second sleeve portion **228**. As shown, the groove **240** is formed in a second sleeve portion outer diameter **244**. The seal **242** may be larger than the groove **240** (e.g., extend radially beyond the second sleeve portion outer diameter **244**) such that the seal **242** is compressed against the bore **110** when the sleeve **202** is installed within the bore **110**. In this matter, fluid may be blocked from entering the first bore section **206**. It should be appreciated that, in alternative configurations, the groove **240** may be omitted and the seal **242** may be moved to act as a face seal that abuts an end of the sleeve **202**, as will be described herein.

[0040] Moving toward the face **214**, a transition **244** is illustrated at the variable diameter portion between the first sleeve portion **226** and the second sleeve portion **228**. While the shelf **234** is formed along the inside, a shoulder **246** is illustrated where the outer diameter changes from the second sleeve portion outer diameter **244** to the first sleeve portion outer diameter **250**. This shoulder **246** may engage an extension **252** between the first bore section **206** and the second bore section **208** representative of the change in diameter, which may help seat or otherwise position the sleeve **202** within the bore **110**. In various embodiments, the shoulder **246** blocks axial movement of the sleeve **202** toward the pressure chamber **112**. As noted above, this may enable operators to determine when the sleeve **202** has been positioned at a desired location. The illustrated embodiment includes the shoulder **246** have a substantially right angle, however, it should be appreciated that the shoulder **246** may be tapered or at a less smaller angle (e.g., between 0 degrees and 90 degrees). Furthermore, in one or more embodiments, the extension **252** may receive a face seal or other seal that is compressed by the sleeve **202**, for example in configurations where the transition **244** is omitted.

[0041] It should be appreciated that various additional seals may also be utilized with the illustrated sleeve **202**. For example, in the illustrated embodiment, additional seal arrangements **254** are arranged surrounding a grease portion **256** and also at an interface with a cap **258**. Accordingly, potential leak paths may be blocked by forming respective grooves that receive seals along various different portions of the sleeve **202**.

[0042] In various embodiments, sleeve **202** may further included threads or coupling components **260**, as illustrated in FIGS. 2A and 2B. In this example, the coupling components **260** extend beyond the face **214**, and in this case are positioned within the cap **258**. In other words, the coupling components **260** may be positioned to not engage the block **104** and/or the bore **110**, but rather, to engage components external to the bore **110**. As noted herein, in various embodiments the external components may correspond to one or more installation/removal tools that may engage the sleeve **202** for efficient removal, for example using internal threads of the tool to engage the coupling components **260** of the sleeve **202**. It should be appreciated that threads are provided as an example for the coupling components **260**, and in various other embodiments, additional fasteners may also be utilized such as tongue and groove fittings, bayonet style connectors, pegs and dowels, glues/adhesives, and the like.

[0043] FIGS. 3A-3C are detailed cross-sectional views of the packing assembly **200** illustrating the sleeve **202** arranged within the bore **110** with the seal **242** positioned within the groove **240** to bear against the block **104** at the second bore section **208**. In other words, the groove **240** and the seal

242 are positioned at the reduced diameter portion of the sleeve **202** corresponding to the second sleeve portion **228**. As noted above, the embodiments shown in FIGS. **3A-3C** are for illustrative purposes and, in various embodiments, the groove **240**, as well as the transition **246**, may be omitted in favor of a substantially constant diameter sleeve **202** that compresses the seal **242** against the fluid end block **104**.

[0044] As described above, the sleeve **202** extends into the bore **110** such that at least a portion of the second sleeve portion **228** is positioned within at least a portion of the second bore section **208**. Further axial movement of the sleeve **202** toward the third bore section **210** may be blocked via engagement between the shoulder **248** and the extension **252**. That is, incorporating variable diameter portions of both the bore **110** and the sleeve **202** may enable operators to quickly and efficiently install components because alignment may be indicated to the operator via contact between components. It should be appreciated that the location of the extension **252** being internal to the block **104** (e.g., within the bore sections) is for illustrative purpose and that, in some instances, the extension **252** may be on the outside of the block **104**.

[0045] The groove **240** is formed within the second sleeve portion **228** and extends a groove depth **300** toward the axis **108** such that a groove diameter **302** is less than a second sleeve portion outer diameter **244**. While the illustrated groove **240** is substantially square or rectangular (e.g., parallel sidewalls with a substantially perpendicular base), it should be appreciated that this arrangement is for illustrative purposes only and that, in other embodiments, the groove **240** may have arcuate sidewalls, slanted sidewalls, a single slanted sidewall, or any other reasonable combination. Furthermore, in one or more embodiments, the groove **240** may be omitted. In various embodiments, a seal thickness **304** may be greater than the groove depth **300**, such that the seal **242** extends radially outward from the groove **240**. As will be appreciated, when installed, the seal **242** may be compressed via contact with the end block **104**, which may cause the seal **242** to expand or otherwise fill different portions of the groove **240**. In various embodiments, the seal **242** is formed from a polymer material, but it should be appreciated that any reasonable material, such as metals, elastomers, synthetics, or combinations thereof may be utilized for the seal **242**. Furthermore, while a single seal **242** is illustrated, multiple seals may be positioned within the groove **240** or multiple grooves **240**.

[0046] The second sleeve portion outer diameter **244** is less than the first sleeve portion outer diameter **250**, as noted above, thereby forming the transition **246** and the shoulder **248**. It should be appreciated that the respective sizes of the diameters **244**, **250** may be particularly selected based on expected operating conditions. For example, the diameters **244**, **250** may be selected for strength, pressure containing purposes, longevity, weight, or the like. As noted, in one or more embodiments the transition **246** and the shoulder **248** may be omitted in favor of a substantially constant outer diameter for the sleeve **202**.

[0047] The packing components **236** are illustrated within the sleeve bore **224** and bear against the plunger **106**. In this embodiment, the packing components **236** include seals, packing brass, and the like. A packing nut **306** secures the packing components **236** within the sleeve bore **224**, for example, via coupling to the block **104**. Accordingly, operation of the pump is enabled as the plunger **106** reciprocates through the bore **110**. It should be appreciated that, in other embodiments, the packing components may not rest on the sleeve **202** or portions thereof and spacer rings, or other sealing elements, may be arranged within the bore **110** to receive the packing components.

[0048] As noted above, additional sealing arrangements **254** are positioned along the sleeve **202**, for example around the grease port **256** and/or proximate the cap **258**. As a result, various levels of leak protection are provided, which may reduce the time between maintenance activities. In at least one embodiment, the additional sealing components may extend toward the cap **258** and stop at the coupling components **260**. For example, the coupling components may be utilized to couple to one or more installation and removal tools utilized during one or more maintenance operations. In at least one embodiment, as noted above, the coupling components **260** are positioned to extend

axially out of the bore **110** and portions may be at least partially overlapped by the cap **258**.

[0049] It should be appreciated that various components of the sleeve **202** may be particularly selected based on operating conditions. For example, a first sleeve portion length **308** is illustrated as being larger than a second sleeve portion length **310** and larger than a first bore section length **312**. That is, in this embodiment, the first sleeve portion **226** extends outwardly from the face **214**. Furthermore, in this configuration, the second sleeve portion length **310** substantially corresponds to a second bore section length **314**. Additionally, due to the variable diameter portions of the sleeve **202**, different portions may have different thicknesses, which may be selected based on the dimensions of associated components. For example, a first sleeve portion thickness **316** is less than a second sleeve portion thickness **318**, which is also less than a transition thickness **320** due to the overlap of the first and second sleeve portions **226**, **228**, at the transition **246**. In various embodiments, relative thicknesses may be substantially constant along the length of the sleeve **202**. [0050] FIG. **4** is an exploded cross-sectional view of the pump assembly **100** including the packing assembly **200**. As shown, the illustrated packing assembly **200** is positioned proximate the face **214** and bore **110**. The seal **242** is positioned out of the groove **240**, formed lower (e.g., closer to the face) than the transition **246**. Furthermore, the packing components **236** and plunger **106** are also installed within the bore, through the sleeve **202**. The packing components **236** are secured in place via the packing nut **306**.

[0051] In operation, the seal **242** may be arranged within the groove **240** and then the sleeve **202** may be installed until the shoulder **248** contacts the extension **252**. In embodiments, coatings or the like may be used to prepare various surfaces. Thereafter, the packing components **236** may be installed, but it should be appreciated that the packing components **236** may also be installed prior to installation of the sleeve **202**. As noted herein, in various embodiments the groove **240** is omitted and the seal **242** may be positioned within the bore **110**, for example on a shelf or spacer ring, and then compressed by the sleeve **202**.

[0052] Further illustrated is the coupling component **260**, which may include threads to facilitate coupling to an installation tool for assembly and/or removal. In this example, the coupling component **260** is positioned at an end of the sleeve **202** extending axially away from the bore **110**. In operation, the coupling component **260** may align with the cap **258**, which in this example is illustrated with a threaded portion **400** over a portion of the cap **258** and a non-threaded portion **402** over another portion of the cap **258**. In various embodiments, the non-threaded portion **402** may align with the coupling component **260**, such that the coupling component **260** is disengaged and/or not coupled to another component during operation. However, the coupling component **260** may be utilized during installation and removal operations. Moreover, in one or more embodiments, the cap **258** may be used to protect or otherwise guard the coupling component **260**. It should be appreciated that, in one or more embodiments, the coupling component **260** may also be configured to couple to the cap **258**.

[0053] FIG. **5** is a cross-sectional view of the sleeve **202** and the seal **242** positioned proximate the bore **110**. As discussed, the groove **240** is formed within the second sleeve portion **228** and has a groove diameter **302** that is less than the second sleeve portion outer diameter **244**. The illustrated seal **242** is sized to fit within the groove **240** and for installation into the bore **110**. However, as noted herein, other embodiments may omit the groove **240** and position the seal **242** at different locations, such as directly into bore **110** axially closer than sleeve **202**. Further illustrated is the transition **246** and the increased transition thickness **320** compared to the first sleeve portion thickness **316** and the second sleeve portion thickness **318**. As noted above, these dimensions may be particularly selected based on anticipated operating conditions. By way of example, it may be desirable for the second sleeve portion **228** to be thicker than the first sleeve portion **226** because the second sleeve portion **226** is closer to the pressure chamber **112** and may experience more exposure to the fluids.

[0054] FIGS. **6A-6F** illustrate an installation and removal sequence **600** in which the coupling

component **260** is utilized for removal and/or installation of one or more sleeves **202**. In this sequence **600**, an installation/removal tool **602** (e.g., tool, installation and removal tool, etc.) is utilized to engage sleeves **202**, for example via coupling components **260**, for removal and/or installation. In one or more embodiments, installation/removal tool **602** may be coupled to one or more components of the pump assembly **100** to facilitate removal, such as via rotational of the pump assembly crankshaft in a forward direction for installation and rotation in a backward direction for removal.

[0055] FIG. **6A** illustrates the pump assembly **100** including the packing assembly **200** having the sleeve **202** as described above. As noted, other embodiments may omit or change various components, but it should be appreciated that one or more steps of the removal and/or installation procedures may be similar. In at least one embodiment, it is determined that removal of the packing assembly **200** is desired, for example to replace the packing components **236**, to inspect the sleeve **202** for damage, or the like. FIG. **6B** further illustrates the pump assembly **100** after the rod **106** is removed from the bore **110**. In this example, the packing nut **306** and cap **258** remain coupled to the block **104**, and moreover, the packing components **236** are still positioned within the bore **110**. FIG. **6C** illustrates removal of the packing nut **306**, which permits removal of the packing components **236**, among other elements. In this example, the packing nut **306** may be threaded into the cap **258** such that removal utilizes unthreading the packing nut **306** from the cap **258** while the cap **258** remains engaged with the block **104**, for example via one or more fasteners coupling the cap **258** to the block **104**.

[0056] Further illustrated in FIG. **6D** is the installation/removal tool **602** positioned proximate the block **104**. In this example, the tool **602** is axially aligned with the axis **108** such that an engaging end **604** is aligned with the sleeve **202**. In one or more embodiments, the engaging end **604** includes mating coupling components **606**, which in this example are threads to engage the coupling components **260** of the sleeve **202**. As shown, an engaging end diameter **608** is greater than the first sleeve portion diameter **230** to facilitate engagement with the coupling components **260**, instead of the packing components **236**. In at least one embodiment, the tool **602** further includes a working end **610**, opposite the engaging end **604**, which includes one or more fittings **612** to facilitate coupling to external drivers, such as the pump assembly **100**, hand tools, etc. to drive rotation of the tool **602**.

[0057] FIG. **6E** illustrates engagement between the tool **602** and the sleeve **202** via the components **260**, **606**. For example, in embodiments where the components **260**, **606** are threads, engagement may be facilitated by rotating the tool **602** about the axis **108**, which in certain embodiments may be performed via coupling of the tool **602** to the pump assembly **100** and rotating the crankshaft. In various embodiments, one or more hand tools may be used to engage the fitting **612** to drive rotation. FIG. **6F** illustrates axial removal of the packing assembly **200** from the bore **110**. In this example, the entire assembly **200** may be removed, which may be an improvement over removing individual components from the bore **110** to facilitate faster installation and removal. Accordingly, a new assembly may be installed, which may have a similar configuration as the removed assembly or may have a different configuration, such as a constant diameter sleeve as described herein.

[0058] FIGS. **7A-7C** are perspective views of embodiments of the pump assembly **100** illustrating different portions of the removal sequence **600**, including removal of the plunger **106** and packing nut **306** (FIG. **7A**), removal of the cap **258** (FIG. **7B**), and installation of the tool **602** (FIG. **7C**). In these examples, various interactions between components may be seen. By way of example, FIG. **7A** illustrates threads for installing the packing nut **306** within the cap **258**. Furthermore, fasteners secure the cap **258** to the block **104**. FIG. **7B** further illustrates removal of the cap **258** to expose the coupling components **260** of the sleeve **202**, which in various embodiments do not engage the cap **258** and are utilized for installation and removal, for example via the tool (not pictured in FIG. **7B**). However, FIG. **7C** illustrates coupling of the tool **602** to the coupling components **260**, for example via mating threads, to facilitate removal of the assembly (not visible). In various

embodiments, it should be appreciated that the tool **602** may further be coupled to one or more components to facilitate rotation, but this has been omitted for clarity with the figures.

[0059] FIG. **8** is a cross-sectional view of an embodiment of a pump assembly **100** including a packing assembly **800**, which may share one or more features with the packing assembly **200**, and it should be appreciated that certain components may share reference numerals for convenience, and such numbering should not be construed as a limitation in configurations utilized with the present disclosure. In this example, the packing assembly **800** includes a sleeve **802**, which may share or more features with the sleeve **202**, as noted above. In one or more embodiments, the sleeve **802** extends, at least partially, into the bore **110**. The sleeve **802** is substantially aligned along the axis **108** and arranged coaxially within the bore **110**. The sleeve **802** is positioned radially outward from the rod **106** and packing components **236** are positioned to bear against both the sleeve **802** and the rod **106**.

[0060] In this example, the sleeve **802** has a substantially constant inner diameter **804** and a substantially constant outer diameter **806**. It should be appreciated that substantially constant refers to a majority percentage of a sleeve length **808** having an approximately equal diameter. In this example, the sleeve **802** is utilized to compress the seal **242** against an inner face **810**. In various embodiments, the inner face **810** is formed, at least in part, on a shelf **812** formed by a transition between diameters within the bore **110**, as also described above. The illustrated seal **242** is positioned at a radially outward portion **814** of the shelf **812** and is substantially aligned with the sleeve **802**. In operation, the sleeve **802** may be utilized to compress the seal **242** to block fluid leaks around the rod **106**.

[0061] In one or more embodiments, a ring **816** is also positioned along the shelf **812**, and in this example is positioned at a radially inward portion **818**. That is, the ring **816** is arranged between the seal **242** and the rod **106**. It should be appreciated that this configuration is for illustrative purposes only and that, in one or more embodiments, at least a portion of the seal may be aligned with the ring **816**, may be radially inward of the ring **816**, or may be in any other reasonable location. In various embodiments, at least a portion of the ring **816** is utilized to support the packing components **236**, as will be described below.

[0062] Further illustrated in FIG. **8** are the coupling components **260**, which as shown as threads. In this example, the coupling components **260** extend axially out of the bore **110** and are overlapped, at least partially, by the cap **258**. In one or more embodiments, the coupling components **260** are not engaged during normal operation of the pump assembly **100** (e.g., not engaged while the pump assembly is running) and are utilized for installation and removal of the packing assembly **900**.

[0063] FIGS. **9A** and **9B** are detailed cross-sectional views of embodiments of the packing assembly **800**. In one or more embodiments, the sleeve **802** is arranged within the bore **110** such to compress the seal **242** against the inner face **810** to block fluid flow from the bore **110**. In at least one embodiment, the sleeve **802** includes substantially constant inner and outer diameters **804**, **806**. In one or more embodiments, the sleeve **802** does not include features to support the packing components **236**, which may be supported by the ring **816** positioned radially inward of the seal **242**.

[0064] FIGS. **9A** and **9B** further illustrate the grease port **256** and seal arrangement **254** described below. As shown, both are axially closer to the pressure chamber **112** (FIG. **1**) than the coupling components **260**, which are shown as threads in the illustrated embodiment. In this example, a coupling component length **900** is shown where a first portion **902** may be considered outside of the bore **110** and a second portion **904** may be considered inside of the bore **110**. It should be appreciated that the portion or percentage of the length **900** which is inside or outside may be particularly selected based on expected operating conditions. Moreover, the component length **900** may be adjusted to accommodate various tools. In this example, it is shown that the coupling component **260** is aligned with the non-threaded portion **402** of the cap **258**, whereas the threaded

portion **400** is utilized to secure the packing nut **306**.

[0065] In this example, the seal **242** is arranged axially closer to the pressure chamber **112** (FIG. 1) than the sleeve **802**. Moreover, the ring **816** is also positioned axially closer than at least a portion of the sleeve **802**. However, an overlapping portion **906** is illustrated between the ring **816** and the sleeve **802**. At the overlapping portion **906**, a sleeve edge **908** is shown to engage a ring edge **910**, where both respective edges **908**, **910** are arranged at angles. As a result, at least a portion of the sleeve edge **908** may have a different diameter than the remainder of the sleeve **802**. A similar configuration is also illustrated at an opposite end of the sleeve **802** and may be utilized to simplify installation and removal, as well as provided a machined surface, among other benefits. It should be appreciated that this configuration is an example and additional angles or mating edges may be utilized, moreover, a curved or arcuate edge may be used to receive one or more of the sleeve **802** and/or the ring **816**. In at least one embodiment, a ring shelf **912** receives and supports the packing components **236**, where the ring shelf **912** is axially further away from the pressure chamber **112** (FIG. 1) than a sleeve end **914**. In other words, the overlapping portion **906** is axially closer to the pressure chamber **112** (FIG. 1) than the packing components **236** in the illustrated embodiment. It should be appreciated that various other configurations may further be utilized within the scope of the present disclosure.

[0066] Various embodiments illustrate both the seal **242** and the ring **816** positioned on the shelf **812**, which is formed at a location within the bore **110** having a changing diameter. It should be appreciated that the seal **242** may be considered a face seal and the shelf **812** may include a machined or sealing surface particularly selected to accommodate a sealing component. A shelf width **916** receives both the seal **242** and the ring **816** and, in certain embodiments, at least a portion of the ring **816** may overlap the shelf **812**.

[0067] In operation, the packing assembly **800** is positioned within the bore **110** such that the sleeve **802** compresses the seal **242**. In one or more embodiments, the seal **242** may be installed within the bore **110** prior to the sleeve **802**. By way of example only, the seal **242** may be installed, followed by the ring **816**, followed by the sleeve **802**, and then the packing components **236**. It should be appreciated that this installation procedure is by way of example only and not intended to limit the scope of the present disclosure.

[0068] FIG. 10 is an exploded cross-sectional view of the pump assembly **100** including the packing assembly **800**. As shown, the illustrated packing assembly **800** is positioned proximate the face **214** and bore **110**. The seal **242** is positioned out of the bore **110** and axially closer to the inner face **810** than the sleeve **802**. Furthermore, the packing components **236** and plunger **106** are also installed within the bore **110**, through the sleeve **802**. The packing components **236** are secured in place via the packing nut **306**.

[0069] In operation, the seal **242** may be arranged within bore **110** against the inner face **810** and then the sleeve **802** may be installed until the sleeve end **914** contacts the seal **242**. In embodiments, coatings or the like may be used to prepare various surfaces. Additionally, in embodiments, the ring **816** is also positioned against the inner face **810**. Thereafter, the packing components **236** may be installed, but it should be appreciated that the packing components **236** may also be installed prior to installation of the sleeve **802**.

[0070] Further illustrated is the coupling component **260**, which may include threads to facilitate coupling to an installation tool for assembly and/or removal. In this example, the coupling component **260** is positioned at an end of the sleeve **802** extending axially away from the bore **110**. In operation, the coupling component **260** may align with the cap **258**, for example with a non-threaded portion (not pictured). In one or more embodiments, the cap **258** may be used to protect or otherwise guard the coupling component **260**. It should be appreciated that, in one or more embodiments, the coupling component **260** may also be configured to couple to the cap **258**. Moreover, while not illustrated in FIG. 10, installation and/or removal may be accomplished using one or more tools, as described herein.

[0071] FIG. 11 is a cross-sectional view of an embodiment of the packing assembly 800 coupled to the installation/removal tool 602. In this example, the mating coupling components 606 of the engaging end 604 are coupled to the coupling components 260 of the sleeve 802 to facilitate installation and removal. As noted above, in various embodiments, the tool 602 may be coupled to one or more components of the assembly to facilitate removal, for example by driving rotation of the tool 602 and/or by driving axial movement of the tool 602.

[0072] FIG. 12 is a flow chart of an embodiment of a method 1200 for installation and removal of a sleeve. It should be appreciated for this method, and all methods described herein, that there may be more or fewer steps. Additionally, the steps may be performed in a different order, or in parallel, unless otherwise specifically stated. In this example, external threads of a first sleeve are engaged via an installation and removal tool 1202. For example, external threads may correspond to a coupling component on a sleeve where the coupling component is not engaged during operation of the pump but engaged during installation and/or removal. Furthermore, it should be appreciated that while threads are discussed with respect to the method, various other coupling mechanisms may be used in other embodiments of the present disclosure. The tool may be coupled to a portion of a pump assembly 1204. In at least one embodiment, the tool is coupled to pump pony rods via a clamp connection, but other locations may also be used. Rotation of a portion of the pump, such as the crankshaft, may be driven in a first direction 1206. In at least one embodiment, rotation axially moves the tool, and thereafter the coupled sleeve, axially away and out of a bore. The first sleeve is disengaged from the tool 1208 and external threads of a second sleeve are engaged 1210. Rotation of the portion of the pump may then be applied in a second, opposite, direction to axially move the second sleeve toward the bore 1212. The tool may then be disengaged from the pump assembly 1214 and the tool may also be disengaged from the second sleeve 1216. In this manner, one or more sleeves may be removed or installed within bores of a pump assembly.

[0073] This application is a divisional of U.S. patent application Ser. No. 17/890,818, filed Aug. 18, 2022, titled “PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL,” which is a divisional of U.S. patent application Ser. No. 17/308,163, filed May 5, 2021, titled “PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL,” now U.S. Pat. No. 11,421,679 issued Aug. 23, 2022, which is a continuation-in-part of U.S. patent application Ser. No. 16/916,657, filed Jun. 30, 2020, titled “SLEEVED PACKING BORE SEALING SYSTEM,” the full disclosures of which are hereby incorporated by reference in their entirety for all purposes.

[0074] The foregoing disclosure and description of the disclosed embodiments is illustrative and explanatory of the embodiments of the disclosure. Various changes in the details of the illustrated embodiments can be made within the scope of the appended claims without departing from the true spirit of the disclosure. The embodiments of the present disclosure should only be limited by the following claims and their legal equivalents.

Claims

1. A packing assembly to be positioned within a bore, comprising: a sleeve to compress a seal against at least a portion of the bore to block fluid flow from the bore, the sleeve positioned to extend along a sleeve length, at least a first portion of the sleeve length positioned within the bore and at least a second portion of the sleeve length positioned to extend axially beyond the bore, the second portion of the sleeve including a coupling component; and packing components positioned radially inward of at least a portion of the sleeve.
2. The packing assembly of claim 1, wherein the coupling component is positioned on a radially outward facing portion of the sleeve.
3. The packing assembly of claim 1, further comprising a ring to be positioned within the bore, the ring to at least partially compress the seal.

4. The packing assembly of claim 3, wherein the packing components are positioned on a ring shelf.

5. The packing assembly of claim 1, wherein the coupling component is not engaged with the bore.

6. A packing assembly to be positioned within a bore, comprising: a sleeve to compress a seal against at least a portion of the bore to block fluid flow from the bore, the sleeve extending along a sleeve length, wherein at least a first portion of the sleeve length is positioned within the bore and at least a second portion of the sleeve length extends axially beyond the bore, the second portion of the sleeve having a coupling component positioned on a radially outward facing portion of the sleeve; a ring positioned within the bore to at least partially compress the seal; and packing components positioned radially inward of at least a portion of the sleeve.

7. The packing assembly of claim 1, wherein the packing components are positioned on a ring shelf.

8. The packing assembly of claim 1, wherein the coupling component is not engaged with the bore.

9. A packing assembly to be positioned within a bore, comprising: a sleeve to compress a seal against at least a portion of the bore to block fluid flow from the bore, the sleeve positioned to extend along a sleeve length, at least a first portion of the sleeve length positioned within the bore and at least a second portion of the sleeve length positioned to extend axially beyond the bore, the second portion of the sleeve including a coupling component; a ring positioned within the bore; and packing components positioned radially inward of at least a portion of the sleeve and located on a ring shelf.

10. The packing assembly of claim 1, wherein the coupling component is positioned on a radially outward facing portion of the sleeve.

11. The packing assembly of claim 1, wherein the ring is positioned to at least partially compress the seal.

12. The packing assembly of claim 1, wherein the coupling component is not engaged with the bore.
